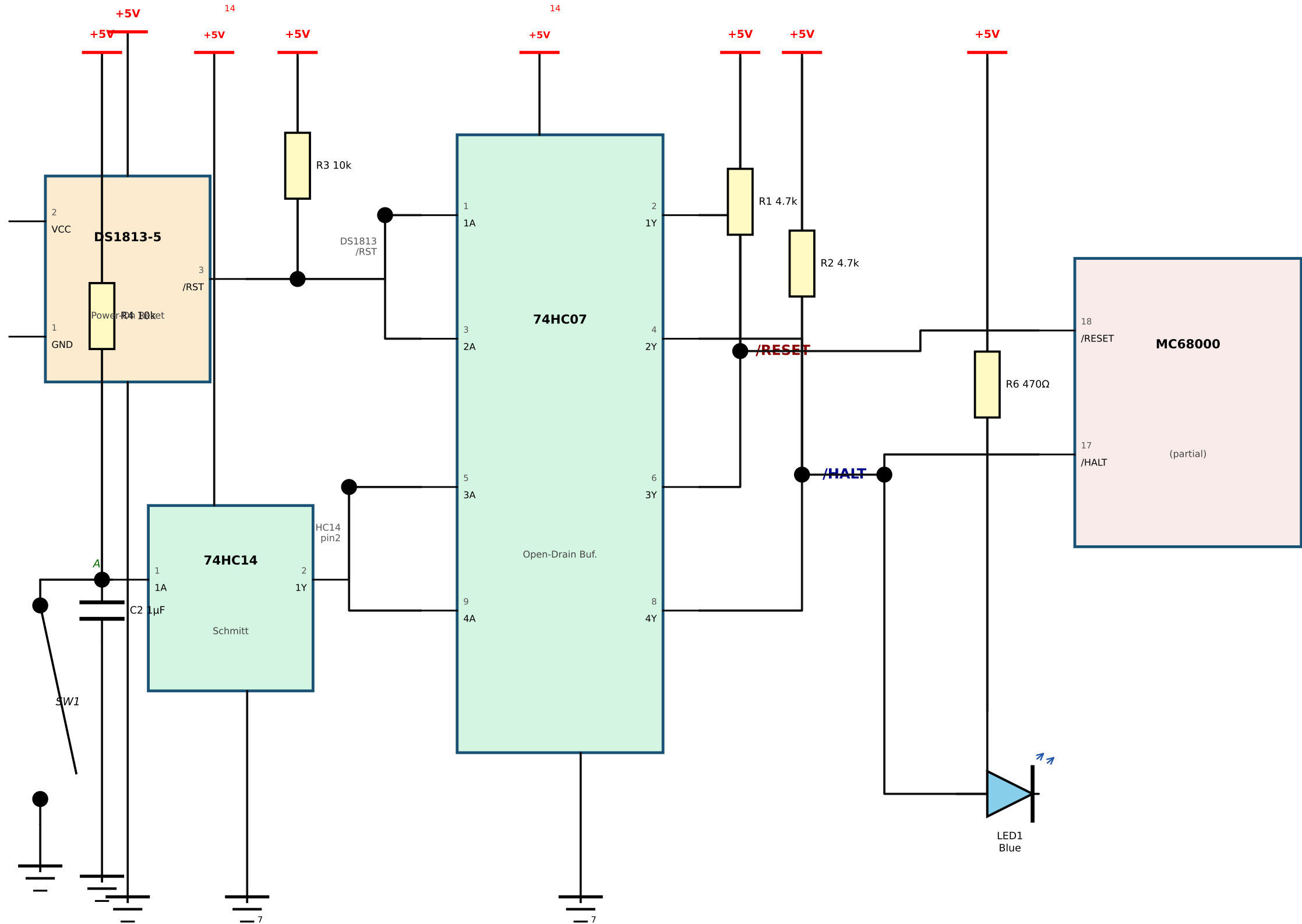


MC68000 /HALT & /RESET Circuit — Schematic



★ Tie all unused 74HC14 / 74HC07 inputs to GND ★ 100nF bypass cap on each IC VCC/GND

MC68000 /HALT & /RESET — Connection Reference

DS1813-5 (SOT-23)

Pin	Name	Connects To
1	GND	GND
2	VCC	+5V
3	/RST	R3 (10kΩ→+5V), HC07 pin1, HC07 pin3

74HC14 (DIP-14) — gate 1 only

Pin	Name	Connects To
1	1A (in)	NODE_A (RC junction)
2	1Y (out)	74HC07 pin5, pin9
7	GND	GND
14	VCC	+5V
3,5,9,11	unused in	GND

74HC07 (DIP-14) — open-drain outputs

Pin	Name	Driven By	Drives	Purpose
1	1A (in)	DS1813 pin3	—	POR input
2	1Y (OD out)	—	/RESET	POR → /RESET
3	2A (in)	DS1813 pin3	—	POR input
4	2Y (OD out)	—	/HALT	POR → /HALT
5	3A (in)	74HC14 pin2	—	Switch input
6	3Y (OD out)	—	/RESET	SW → /RESET
7	GND	GND	—	
8	4Y (OD out)	—	/HALT	SW → /HALT
9	4A (in)	74HC14 pin2	—	Switch input
11,13	5A,6A (in)	GND	—	Tie unused
14	VCC	+5V	—	

MC68000 (relevant pins only)

Pin	Name	Connects To
14	VCC	+5V
16	GND	GND
17	/HALT	R2(4.7k→+5V), HC07 pin4,pin8, LED cathode
18	/RESET	R1(4.7k→+5V), HC07 pin2,pin6

NODE_A — Debounce Junction

Part	Value	Connects To
R4	10kΩ	+5V
C2	1μF	GND
SW1	SPST-NO	GND (other terminal)
74HC14 pin1	—	HC14 1A input

MC68000 /HALT & /RESET — BOM & Behaviour

Bill of Materials

Ref	Part	Value / Part#	Notes
R1, R2	Resistor	4.7kΩ	/RESET and /HALT pull-ups
R3	Resistor	10kΩ	DS1813 /RST output pull-up
R4	Resistor	10kΩ	SW1 pull-up (NODE_A)
R6	Resistor	470Ω	LED current limit (~3.8mA)
C2	Capacitor	1μF	SW1 debounce (τ = 10ms)
C3,C4,C5	Capacitor	100nF ceramic	Bypass, one per IC VCC/GND
U1	74HC14	DIP-14	Hex Schmitt inverter (1 gate used)
U2	74HC07	DIP-14	Hex open-drain buffer (4 gates used)
U3	DS1813-5	SOT-23	Voltage supervisor, 150ms reset pulse
SW1	Push button	SPST-NO	Panel mount momentary
LED1	Blue LED	3mm or 5mm	Vf ≈ 3.2V

Behaviour Summary

Event	/RESET	/HALT	LED	Result
Power-on	LOW	LOW	ON	CPU resets (DS1813, 150ms)
Brownout < 4.65V	LOW	LOW	ON	CPU resets (DS1813)
SW1 pressed	LOW	LOW	ON	CPU resets (manual)
CPU executes RESET opcode	LOW*	HIGH	OFF	Peripherals reset, CPU runs
CPU double-bus-fault halt	HIGH	LOW	ON	CPU halted, blue LED on
Normal operation	HIGH	HIGH	OFF	Running normally

* CPU drives /RESET LOW itself. Because /RESET and /HALT are isolated (separate HC07 gate outputs per pin), /HALT stays HIGH — CPU does not halt.